

Product Requirements Document (PRD)

Project Title

GEO/AIO Search Analytics Engine

Project Owner

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Objective

Build a system that measures Peomiz's visibility across AI-powered search platforms such as ChatGPT, Gemini, and Claude by collecting AI responses, detecting brand mentions, analyzing recommendation positions, and generating actionable GEO insights.

Business Goals

- Measure AI search visibility
 - Benchmark competitors
 - Track recommendation frequency
 - Support Generative Engine Optimization (GEO)
 - Create a centralized reporting framework
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Module 1: Prompt Management System

Create structured prompt categories.

Service-Based Prompts

- Best appliance repair service in Mumbai
- Best home cleaning service in Bangalore
- Best AC service near me

Hyperlocal Prompts

- Best appliance repair service in Kandivali East
- Home cleaning services in Andheri West
- Electrician services in Whitefield

Competitive Prompts

- Urban Company alternatives
- Best home service providers in India

Stored Fields:

- Prompt ID
 - Prompt Category
 - Service Type
 - Location
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Module 2: Automated LLM Query Pipeline

Supported Models:

- OpenAI (ChatGPT)
- Google Gemini
- Anthropic Claude

System Functions:

- Execute prompt batches
- Store responses
- Retry failed requests
- Maintain execution logs
- Monitor API failures

Captured Data:

- Prompt
- Model
- Response
- Timestamp

Operational Constraints

- Prompt batching to reduce API costs
 - Retry mechanism for transient failures
 - Query volume controlled according to budget limits
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Module 3: Mention Detection Engine

Brand Detection

Detect:

- Peomiz
- Competitor names

Output:

- Mentioned (Yes/No)
- Competitor Mentioned (Yes/No)

Position Detection

Structured Responses

Example:

1. Urban Company
2. Peomiz
3. Housejoy

Output:

Peomiz Rank = 2

Unstructured Responses

Example:

"Urban Company is widely used. Peomiz is also emerging as a strong alternative."

Output:

Mention Position:

- Early
- Middle
- Late

Sentiment Analysis (MVP)

The MVP will use a lightweight rule-based sentiment approach based on keyword and polarity matching.

Output:

- Positive
- Neutral
- Negative

Advanced ML-based sentiment classification is considered a future enhancement.

Module 4: Visibility Scoring System

Purpose:

Generate a comparable visibility score across different models and prompts.

Formula

Visibility Score =
Mention Weight × Position Weight × Sentiment Weight

Mention Weight

- Mentioned = 1
- Not Mentioned = 0

Position Weight (Structured Responses)

- Rank 1 = 5

- Rank 2 = 4
- Rank 3 = 3

Position Weight (Unstructured Responses)

- Early Mention = 3
- Middle Mention = 2
- Late Mention = 1

Sentiment Weight

- Positive = 1.2
- Neutral = 1.0
- Negative = 0.8

Scoring Note

Structured ranking and unstructured mention placement use separate evaluation methodologies and are normalized independently. These scores should not be interpreted as equivalent ranking confidence levels.

Module 5: Database Layer

Database:

PostgreSQL

Core Tables:

- prompts
- responses
- mentions
- visibility_scores

Stored Metrics:

- Date
- Prompt
- LLM
- Mention Status
- Position
- Sentiment

- Visibility Score
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Module 6: Analytics Dashboard (MVP)

Visibility Metrics

- Total Mentions
- Mention Rate
- Visibility Score

Competitive Analysis

- Competitor Mentions
- Relative Position

Trend Analysis

- Daily Visibility
- Weekly Visibility
- Model-wise Visibility

GEO Insights

- Top Performing Prompts
 - Visibility Changes
 - Optimization Opportunities
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Technical Stack

Backend

- Python
- FastAPI

Database

- PostgreSQL

Processing Layer

- Pandas
- Regex
- Basic NLP Libraries

Dashboard

- Streamlit (MVP)

Future Phase

- React + FastAPI Dashboard

AI Integrations

- OpenAI API
 - Google Gemini API
 - Anthropic Claude API
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Risks & Assumptions

1. LLM responses may vary significantly in structure.
 2. Exact ranking extraction may not always be possible.
 3. API rate limits and usage costs may restrict query volume.
 4. Visibility scoring methodology may evolve after initial testing.
 5. Initial sentiment analysis will be rule-based and lightweight.
 6. Daily prompt volume may need adjustment based on API budget constraints.
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Out of Scope

- Custom LLM Training
- Proprietary AI Models
- Customer Churn Analytics
- Internal Product Analytics
- Paid Advertising Analytics
- Full Enterprise SaaS Platform

Timeline

Week 1

Research GEO/AIO ecosystem
Create prompt framework

Week 2

OpenAI & Gemini integration
Response collection pipeline

Week 3

Claude integration
Database schema
Mention detection engine

Week 4

Position analysis
Visibility scoring
Data validation

Week 5

Dashboard development
Reporting layer

Week 6

Testing
Documentation
Final demo

Final Deliverables

- Automated Query Pipeline
- Mention Detection Engine
- Visibility Scoring System
- PostgreSQL Database
- Analytics Dashboard MVP
- Technical Documentation
- GEO Optimization Recommendation Report